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Inventor(s)

Chary; Indra V. et al.

MICROELECTRONIC DEVICES AND MEMORY DEVICES

Abstract

A microelectronic device comprises a stack structure, contact structures, and additional contact structures. The stack structure comprises a vertically alternating sequence of conductive material and insulative material arranged in tiers. The stack structure is divided into blocks each comprising a stadium structure including steps comprising horizontal ends of the tiers. The contact structures are within a horizontal area of the stadium structure and vertically extend through the stack structure. The additional contact structures are on at least some of the steps of the stadium structure and are coupled to the contact structures. Memory devices and electronic devices are also disclosed.

Inventors: Chary; Indra V. (Boise, ID), Luo; Shuangqiang (Boise, ID), Xu; Lifang (Boise, ID)

Applicant: Micron Technology, Inc. (Boise, ID)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/349,158, filed Jun. 16, 2021, which will issue as U.S. Pat. No. 12,315,801 on May 27, 2025, the disclosure of which is hereby incorporated herein in its entirety by this reference.

TECHNICAL FIELD

[0002] The disclosure, in various embodiments, relates generally to the field of microelectronic device design and fabrication. More specifically, the disclosure relates to microelectronic devices including contact structures within a horizontal area of a stadium structure, and to related electronic systems and methods.

BACKGROUND

[0003] A continuing goal of the microelectronics industry has been to increase the memory density (e.g., the number of memory cells per memory die) of memory devices, such as non-volatile memory devices (e.g., NAND Flash memory devices). One way of increasing memory density in non-volatile memory devices is to utilize vertical memory array (also referred to as a “three-dimensional (3D) memory array”) architectures. A conventional vertical memory array includes vertical memory strings extending through openings in one or more deck(s) (e.g., stack structure(s)) including tiers of conductive material and insulative material. Each vertical memory string may include at least one select device coupled in series to a serial combination of vertically stacked memory cells. Such a configuration permits a greater number of switching devices (e.g., transistors) to be located in a unit of die area (i.e., length and width of active surface consumed) by building the array upwards (e.g., vertically) on a die, as compared to structures with conventional planar (e.g., two-dimensional) arrangements of transistors.

[0004] Vertical memory array architectures generally include electrical connections between the conductive structures of the tiers of the deck(s) control logic devices (e.g., string drivers) so that the memory cells of the vertical memory array can be uniquely selected for writing, reading, or erasing operations. One method of forming such an electrical connection includes forming so-called “staircase” (or “stair step”) structures at edges (e.g., horizontal ends) of the tiers of the deck(s) of the memory device. The staircase structure includes individual “steps” defining contact regions for the conductive material of the tiers, upon which conductive contact structures can be positioned to provide electrical access to the conductive material. In turn, additional conductive structures (e.g., additional conductive contact structures, conductive routing structures) can be employed to couple the conductive contact structures to the control logic devices. However, the additional conductive structures can occupy valuable horizontal space within the memory device that can limit device performance enhancements and/or size reductions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is simplified, partial perspective view illustrating a microelectronic device structure of a microelectronic device, in accordance with embodiments of the disclosure;

[0006] FIG. 2A is a simplified, longitudinal cross-sectional view of a portion A of the microelectronic device structure shown in FIG. 1. FIG. 2B is a simplified, partial longitudinal cross-sectional view of a portion of the microelectronic device structure shown in FIG. 2A about the dashed line B-B shown in FIG. 2A. FIG. 2C is a simplified, partial plan view of the portion A of the microelectronic device structure shown in FIG. 1;

[0007] FIGS. 3A, 3B, and 3C are simplified, partial plan views depicting different contact configuration options within the portion A of the microelectronic device structure shown in FIG. 1, in accordance with additional embodiments of the disclosure;

[0008] FIGS. 4A, 4B, and 4C are simplified, partial plan views depicting different contact and routing configuration options within the portion A of the microelectronic device structure shown in FIG. 1, in accordance with embodiments of the disclosure;

[0009] FIG. 5 is a simplified, partial cutaway perspective view of a microelectronic device, in accordance with embodiments of the disclosure; and

[0010] FIG. 6 is a schematic block diagram illustrating an electronic system, in accordance with embodiments of the disclosure.

DETAILED DESCRIPTION

[0011] The illustrations included herewith are not meant to be actual views of any particular systems, microelectronic structures, microelectronic devices, or integrated circuits thereof, but are merely idealized representations that are employed to describe embodiments herein. Elements and features common between figures may retain the same numerical designation except that, for ease of following the description, reference numerals begin with the number of the drawing on which the elements are introduced or most fully described.

[0012] The following description provides specific details, such as material types, material thicknesses, and processing conditions in order to provide a thorough description of embodiments described herein. However, a person of ordinary skill in the art will understand that the embodiments disclosed herein may be practiced without employing these specific details. Indeed, the embodiments may be practiced in conjunction with conventional fabrication techniques employed in the semiconductor industry. In addition, the description provided herein does not form a complete process flow for manufacturing a microelectronic device (e.g., a semiconductor device, a memory device, such as NAND Flash memory device), apparatus, or electronic system, or a complete microelectronic device, apparatus, or electronic system. The structures described below do not form a complete microelectronic device, apparatus, or electronic system. Only those process acts and structures necessary to understand the embodiments described herein are described in detail below. Additional acts to form a complete microelectronic device, apparatus, or electronic system from the structures may be performed by conventional techniques.

[0013] Unless the context indicates otherwise, the materials described herein may be formed by any suitable technique including, but not limited to, spin coating, blanket coating, chemical vapor deposition (CVD), plasma enhanced CVD (PECVD), atomic layer deposition (ALD), plasma enhanced ALD (PEALD), physical vapor deposition (PVD) (e.g., sputtering), or epitaxial growth. Depending on the specific material to be formed, the technique for depositing or growing the material may be selected by a person of ordinary skill in the art. In addition, unless the context indicates otherwise, removal of materials described herein may be accomplished by any suitable technique including, but not limited to, etching (e.g., dry etching, wet etching, vapor etching), ion milling, abrasive planarization (e.g., chemical-mechanical planarization (CMP)), or other known methods.

[0014] As used herein, the terms “longitudinal,” “vertical,” “lateral,” and “horizontal” are in reference to a major plane of a structure are not necessarily defined by Earth's gravitational field. A “horizontal” or “lateral” direction is a direction that is substantially parallel to the major plane of the structure, while a “vertical” or “longitudinal” direction is a direction that is substantially perpendicular to the major plane of the structure. The major plane of the structure is defined by a

surface of the structure having a relatively large area compared to other surfaces of the structure. With reference to the figures, a “horizontal” or “lateral” direction may be perpendicular to an indicated “Z” axis, and may be parallel to an indicated “X” axis and/or parallel to an indicated “Y” axis; and a “vertical” or “longitudinal” direction may be parallel to an indicated “Z” axis, may be perpendicular to an indicated “X” axis, and may be perpendicular to an indicated “Y” axis.

[0015] As used herein, the term “substantially” in reference to a given parameter, property, or condition means and includes to a degree that one of ordinary skill in the art would understand that the given parameter, property, or condition is met with a degree of variance, such as within acceptable tolerances. By way of example, depending on the particular parameter, property, or condition that is substantially met, the parameter, property, or condition may be at least 90.0 percent met, at least 95.0 percent met, at least 99.0 percent met, at least 99.9 percent met, or even 100.0 percent met.

[0016] As used herein, “about” or “approximately” in reference to a numerical value for a particular parameter is inclusive of the numerical value and a degree of variance from the numerical value that one of ordinary skill in the art would understand is within acceptable tolerances for the particular parameter. For example, “about” or “approximately” in reference to a numerical value may include additional numerical values within a range of from 90.0 percent to 110.0 percent of the numerical value, such as within a range of from 95.0 percent to 105.0 percent of the numerical value, within a range of from 97.5 percent to 102.5 percent of the numerical value, within a range of from 99.0 percent to 101.0 percent of the numerical value, within a range of from 99.5 percent to 100.5 percent of the numerical value, or within a range of from 99.9 percent to 100.1 percent of the numerical value.

[0017] As used herein, spatially relative terms, such as “beneath,” “below,” “lower,” “bottom,” “above,” “upper,” “top,” “front,” “rear,” “left,” “right,” and the like, may be used for ease of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Unless otherwise specified, the spatially relative terms are intended to encompass different orientations of the materials in addition to the orientation depicted in the figures. For example, if materials in the figures are inverted, elements described as “below” or “beneath” or “under” or “on bottom of” other elements or features would then be oriented “above” or “on top of” the other elements or features. Thus, the term “below” can encompass both an orientation of above and below, depending on the context in which the term is used, which will be evident to one of ordinary skill in the art. The materials may be otherwise oriented (e.g., rotated 90 degrees, inverted, flipped, etc.) and the spatially relative descriptors used herein interpreted accordingly.

[0018] As used herein, features (e.g., regions, materials, structures, devices) described as “neighboring” one another means and includes features of the disclosed identity (or identities) that are located most proximate (e.g., closest to) one another. Additional features (e.g., additional regions, additional materials, additional structures, additional devices) not matching the disclosed identity (or identities) of the “neighboring” features may be disposed between the “neighboring” features. Put another way, the “neighboring” features may be positioned directly adjacent one another, such that no other feature intervenes between the “neighboring” features; or the “neighboring” features may be positioned indirectly adjacent one another, such that at least one feature having an identity other than that associated with at least one the “neighboring” features is positioned between the “neighboring” features. Accordingly, features described as “vertically neighboring” one another means and includes features of the disclosed identity (or identities) that are located most vertically proximate (e.g., vertically closest to) one another. Moreover, features described as “horizontally neighboring” one another means and includes features of the disclosed identity (or identities) that are located most horizontally proximate (e.g., horizontally closest to) one another.

[0019] As used herein, the term “memory device” means and includes a microelectronic device

exhibiting memory functionality, but not necessarily limited to memory functionality. Stated another way, and by way of example only, the term “memory device” means and includes not only conventional memory (e.g., conventional volatile memory, such as conventional dynamic random access memory (DRAM); conventional non-volatile memory, such as conventional NAND memory), but also includes an application specific integrated circuit (ASIC) (e.g., a system on a chip (SoC)), a microelectronic device combining logic and memory, and a graphics processing unit (GPU) incorporating memory.

[0020] As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0021] As used herein, “and/or” includes any and all combinations of one or more of the associated listed items.

[0022] As used herein, the phrase “coupled to” refers to structures operatively connected with each other, such as electrically connected through a direct Ohmic connection or through an indirect connection (e.g., by way of another structure).

[0023] As used herein, “conductive material” means and includes electrically conductive material such as one or more of a metal (e.g., tungsten (W), titanium (Ti), molybdenum (Mo), niobium (Nb), vanadium (V), hafnium (Hf), tantalum (Ta), chromium (Cr), zirconium (Zr), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Ir), nickel (Ni), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag), gold (Au), aluminum (Al)), an alloy (e.g., a Co-based alloy, an Fe-based alloy, an Ni-based alloy, an Fe- and Ni-based alloy, a Co- and Ni-based alloy, an Fe- and Co-based alloy, a Co- and Ni- and Fe-based alloy, an Al-based alloy, a Cu-based alloy, a magnesium (Mg)-based alloy, a Ti-based alloy, a steel, a low-carbon steel, a stainless steel), a conductive metal-containing material (e.g., a conductive metal nitride, a conductive metal silicide, a conductive metal carbide, a conductive metal oxide), and a conductively doped semiconductor material (e.g., conductively doped polysilicon, conductively doped germanium (Ge), conductively doped silicon germanium (SiGe)). In addition, a “conductive structure” means and includes a structure formed of and including conductive material.

[0024] As used herein, “insulative material” means and includes electrically insulative material, such as one or more of at least one dielectric oxide material (e.g., one or more of a silicon oxide (SiO.sub.x), phosphosilicate glass, borosilicate glass, borophosphosilicate glass, fluorosilicate glass, an aluminum oxide (AlO.sub.x), a hafnium oxide (HfO.sub.x), a niobium oxide (NbO.sub.x), a titanium oxide (TiO.sub.x), a zirconium oxide (ZrO.sub.x), a tantalum oxide (TaO.sub.x), and a magnesium oxide (MgO.sub.x)), at least one dielectric nitride material (e.g., a silicon nitride (SiN.sub.y)), at least one dielectric oxynitride material (e.g., a silicon oxynitride (SiO.sub.xN.sub.y)), and at least one dielectric carboxynitride material (e.g., a silicon carboxynitride (SiO.sub.xC.sub.zN.sub.y)). Formulae including one or more of “x,” “y,” and “z” herein (e.g., SiO.sub.x, AlO.sub.x, HfO.sub.x, NbO.sub.x, TiO.sub.x, SiN.sub.y, SiO.sub.xN.sub.y, SiO.sub.xC.sub.zN.sub.y) represent a material that contains an average ratio of “x” atoms of one element, “y” atoms of another element, and “z” atoms of an additional element (if any) for every one atom of another element (e.g., Si, Al, Hf, Nb, Ti). As the formulae are representative of relative atomic ratios and not strict chemical structure, an insulative material may comprise one or more stoichiometric compounds and/or one or more non-stoichiometric compounds, and values of “x,” “y,” and “z” (if any) may be integers or may be non-integers. As used herein, the term “non-stoichiometric compound” means and includes a chemical compound with an elemental composition that cannot be represented by a ratio of well-defined natural numbers and is in violation of the law of definite proportions. In addition, an “insulative structure” means and includes a structure formed of and including insulative material.

[0025] As used herein, the term “homogeneous” means relative amounts of elements included in a feature (e.g., a material, a structure) do not vary throughout different portions (e.g., different horizontal portions, different vertical portions) of the feature. Conversely, as used herein, the term

“heterogeneous” means relative amounts of elements included in a feature (e.g., a material, a structure) vary throughout different portions of the feature. If a feature is heterogeneous, amounts of one or more elements included in the feature may vary stepwise (e.g., change abruptly), or may vary continuously (e.g., change progressively, such as linearly, parabolically) throughout different portions of the feature. The feature may, for example, be formed of and include a stack of at least two different materials.

[0026] FIG. 1 is simplified, partial perspective view illustrating a microelectronic device structure **100** of a microelectronic device (e.g., a memory device, such as a 3D NAND Flash memory device), in accordance with embodiments of the disclosure. The microelectronic device structure **100** includes a stack structure **102** including conductive material **134** vertically alternating with insulative material **104**. As shown in FIG. 1, the stack structure **102** may have tiers **108** each including a level of the conductive material **134** vertically neighboring a level of the insulative material **104**. In some embodiments, the insulative material **104** is formed of and includes dielectric oxide material (e.g., SiO.sub.x such as silicon dioxide (SiO.sub.2)), and the conductive material **134** is formed of and includes W.

[0027] As shown in FIG. 1, the stack structure **102** may be divided into blocks **110** separated by slot structures **142**. Each block **110** may include a stadium region made up of multiple stadium structures **114**. For example, an individual block **110** may include a first stadium structure **114A**, a second stadium structure **114B**, a third stadium structure **114C**, and a fourth stadium structure **114D**. The second stadium structure **114B** may be positioned relatively vertically lower (e.g., in the Z-direction) within the block **110** than the first stadium structure **114A**; the third stadium structure **114C** may be positioned relatively vertically lower within the block **110** than the second stadium structure **114B**; and the fourth stadium structure **114D** may be positioned relatively vertically lower within the block **110** than the third stadium structure **114C**.

[0028] Within each block **110** of the stack structure **102**, the conductive material **134** of one or more relatively vertically higher tier(s) **108** (e.g., upper tiers) may be employed to form upper select gate structures (e.g., drain side select gate (SGD) structures) for upper select transistors (e.g., drain side select transistors) of the block **110**. The conductive material **134** of the relatively vertically higher tier(s) **108** may be segmented by one or more filled slot(s) (e.g., filled SGD slot(s)) to form the upper select gate structures of the block **110**. In some embodiments, within each block **110** of the stack structure **102**, the conductive material **134** of each of less than or equal to eight (8) relatively higher tier(s) **108** (e.g., from one (1) relatively vertically higher tier **108** to eight (8) relatively vertically higher tiers **108**) of the stack structure **102** is employed to form upper select gate structures (e.g., SGD structures) for the block **110**. In addition, within each block **110** of the stack structure **102**, the conductive material **134** of at least some relatively vertically lower tiers **108** vertically underlying the relatively vertically higher tier(s) **108** may be employed to form access line structures (e.g., word line structures) of the block **110**, as also described in further detail below. Moreover, within each block **110** of the stack structure **102**, the conductive material **134** of at least a vertically lowest tier **108** may be employed to form as at least one lower select gate structure (e.g., at least one source side select gate (SGS) structure) for lower select transistors (e.g., source side select transistors) of the block **110**.

[0029] The stadium structures **114** may individually include staircase structures **116** defined by horizontal ends (e.g., edges) of the tiers **108** of the stack structure **102**. Each stadium structure **114** may, for example, include a forward staircase structure **116A** exhibiting positive slope, a reverse staircase structure **116B** exhibiting negative slope, and a central region **117** horizontally interposed between the forward staircase structure **116A** and the reverse staircase structure **116B**. Each staircase structure **116** may include steps **118** defined by horizontal ends of at least some of the tiers **108**. In addition, openings partially defined by the stadium structures **114** may be filled with dielectric material so as to form filled trenches **120** (seen in more detail in FIGS. 2A and 2B).

[0030] Still referring to FIG. 1, each block **110** of the stack structure **102** further includes crest

regions **122** (which may also be referred to as “elevated regions” or “plateau regions”) horizontally interposed (e.g., in the X-direction) between horizontally neighboring stadium structures **114**. The crest regions **122** may define horizontal boundaries (e.g., in the X-direction and in the Y-direction) of unremoved portions of the tiers **108** of the stack structure **102**. As described in further detail below, the crest regions **122** may individual exhibit reduced horizontal dimensions (e.g., in the X-direction) (and, hence, reduced horizontal areas) as compared to the crest regions employed within conventional microelectronic devices. In some embodiments, individual crest regions **122** of individual blocks **110** of the stack structure **102** exhibit a width in the X-direction of less than about 1 micrometer (μm), such as less than or equal to about 500 nanometers (nm), or less than or equal to about 100 nm.

[0031] As described in further detail below with reference to FIGS. 2A through 2C, within horizontal areas of the stadium structures **114**, individual blocks **110** of the stack structure **102** further include contact structures. For example, within the horizontal area of an individual stadium structure **114**, first contact structures may vertically extend through the stack structure **102** and may be coupled to control logic circuitry underlying the stack structure **102**; second contact structures may vertically extend through the stack structure **102** and may be electrically isolated from the control logic circuitry; and third contact structures (e.g., word line contact structures) may contact (e.g., physically contact, land on) the conductive material **134** of the tiers **108** of the stack structure **102**, and may be coupled to the first contact structures by way of additional conductive structures (e.g., additional contact structures, routing structures). The inclusion of the first contact structures within the horizontal areas of the stadium structures **114** may facilitate reductions in horizontal dimensions (e.g., in the X-direction) of the crest regions **122** of the blocks **110** of the stack structure **102** as compared to conventional configurations including contact structures functionally corresponding to the first contact structures within horizontal areas of the crest regions thereof. As described in further detail below, individual steps **118** of the staircase structures **116** of an individual stadium structure **114** may have a quantity of contact structures (e.g., first contact structures, second contact structures, third contact structures) associated therewith (e.g., at least partially with horizontal boundaries thereof).

[0032] Still referring to FIG. 1, within an individual block **110** of the stack structure **102**, the first stadium structure **114A** may exhibit different contact structures and/or a different arrangement of contact structures within a horizontal area thereof than at least some other of the stadium structures **114** (e.g., the second stadium structure **114B**, the third stadium structure **114C**, the fourth stadium structure **114D**) of the block **110**. For example, the first stadium structure **114A** may have a reduced number of at least the first contact structures within horizontal boundaries thereof as compared to one or more other of the stadium structures **114**. The reduced number of the first contact structures may be associated with the use of the first stadium structure **114A** as a so-called SGD stadium, and with the positions of the one or more filled slot(s) (e.g., filled SGD slot(s)) employed to form the upper select gate structures (e.g., SGD structures) of the block **110**. At least some of the first contact structures associated with the first stadium structure **114A** may instead be positioned within a horizontal area of at least one of the crest regions **122** horizontally neighboring the first stadium structure **114A**, which may result these crest region(s) **122** having relatively larger horizontal areas than other crest regions **122** within the block **110**. In some embodiments, the first stadium structure **114A** is free of first contact structures within the horizontal area thereof. In additional embodiments, the first stadium structure **114A** includes some first contact structures within horizontal boundaries thereof.

[0033] FIG. 2A is a simplified, longitudinal cross-sectional view of portion A (identified with a dashed box in FIG. 1) of the microelectronic device structure **100** depicted in FIG. 1. The portion A encompasses the second stadium structure **114B** of an individual block **110** (FIG. 1) of the stack structure **102** (FIG. 1). The portion A also encompasses parts of the crest regions **122** horizontally neighboring the second stadium structure **114B** in the X-direction; and the filled trench **120** having

boundaries defined by the second stadium structure **114B**. While additional features (e.g., structures, materials) of the microelectronic device structure **100** are described hereinbelow with reference to the portion A of the microelectronic device structure **100**, such additional features may also be formed and included in additional portions of the microelectronic device structure **100**, including additional portions encompassing additional stadium structures **114** of one or more (e.g., each) of the blocks **110** (FIG. **1**) of the stack structure **102** (FIG. **1**) and parts of the crest regions **122**, and the filled trenches **120** having boundaries defined by the additional stadium structures **114**. FIG. **2B** is a simplified, partial longitudinal cross-sectional view of a portion of the microelectronic device structure **100** of FIGS. **1** and **2A** about a dashed line B-B shown in FIG. **2A**. FIG. **2C** is a simplified, partial plan of the portion A of microelectronic device structure **100** depicted in FIG. **1**.

[0034] Referring collective to FIGS. **2A** through **2C**, within a horizontal area of the second stadium structure **114B**, first contact structures **140** and second contact structures **141** (FIG. **2B**) are formed to individually vertically extend through the stack structure **102**, and third contact structures **165** are formed to land on the conductive material **134** of the tiers **108** at the steps **118** of the second stadium structure **114B**. In addition, as shown in FIG. **2B**, the first contact structures **140** may be coupled to the third contact structures **165** (and, hence, the conductive material **134** of the tiers **108**) by way of first additional contact structures **124**, routing structures **146**, and second additional contact structures **125**. The first additional contact structures **124** may be positioned on or over the first contact structures **140**, the second additional contact structures **125** may be positioned on or over the third contact structures **165**, and the routing structures **146** may horizontally extend from and between the first additional contact structures **124** and the second additional contact structures **125**. The first contact structures **140** may also be coupled to control logic circuitry (e.g., string driver circuitry) vertically underlying the stack structure **102**, as described in further detail below. In some embodiments, within an individual block **110**, the first contact structures **140** are positioned relatively more proximate the filled slot structures **112** than the second contact structures **141**. The second contact structures **141** are not coupled to the third contact structures **165**. In FIG. **2B**, the third contact structures **165** are depicted with dashed lines to indicate that the third contact structure **165** are horizontally offset from (e.g., in the X-direction) first contact structures **140** and the second contact structures **141**. The first contact structures **140**, the second contact structures **141**, and the third contact structures **165** vertically extend through the filled trench **120** vertically overlying the second stadium structure **114B**.

[0035] The first contact structures **140**, the second contact structures **141**, and the third contact structures **165**, the first additional contact structures **124**, the second additional contact structures **125**, and the routing structures **146** may individually be formed of and include conductive material. In some embodiments, the first contact structures **140**, the second contact structures **141**, and the third contact structures **165**, the first additional contact structures **124**, the second additional contact structures **125**, and the routing structures **146** are formed of and include one or more of W and TiN. Subsequently, dielectric liners (e.g., dielectric oxide liners) may be formed to be horizontally interposed between at least the conductive material **134** of the tiers **108** of the stack structure **102** and the first contact structures **140** and the second contact structures **141** (if any). Each of the dielectric liners may individually have a horizontal thickness with a range of from about 90 nm to about 120 nm, such as about 100 nm.

[0036] The first contact structures **140**, the second contact structures **141**, and the third contact structures **165** may be positioned at desirable locations relative to one another and the steps **118** of the staircase structures **116**. Referring to FIG. **2C**, in some embodiments, each third contact structure **165** is individually substantially horizontally centrally (e.g., in the X-direction, in the Y-direction) positioned on one of the steps **118** of the staircase structures **116**. The first contact structures **140** and the second contact structure **141** may be horizontally offset from the third contact structures **165**. An individual step **118** having one of the third contact structures **165** within

horizontal boundaries thereof may also exhibit one or more of the first contact structures **140** at least partially within the horizontal boundaries thereof, and, optionally, one or more of the second contact structures **141** at least partially within the horizontal boundaries thereof. In some embodiments, columns individually including one or more (e.g., two or more) of the first contact structures **140** and, optionally, one or more (e.g., one) of the second contact structures **141** horizontally alternate with the third contact structures **165** in the X-direction and horizontally extend in the Y-direction. At least some of the first contact structures **140** within an individual column may be positioned within horizontal boundaries of the same step **118** as one another. Within each column, the first contact structures **140** and the second contact structures **141** (if any) thereof may be substantially aligned with one another in the X-direction. In addition, in some embodiments, rows individually including some (e.g., two or more) of the first contact structures **140** and, optionally, some (e.g., one or more) of the second contact structures **141** may horizontally extend in the Y-direction. Within each row, the first contact structures **140** and the second contact structures **141** (if any) thereof may be substantially aligned with one another in the Y-direction. At least some of the first contact structures **140** within an individual row may be at least partially positioned within horizontal boundaries of different steps **118** than one another.

[0037] As shown in FIG. 2C, in some embodiments, within horizontal boundaries of one of the staircase structures **116**, one (1) row of the third contact structures **165** is interposed in the Y-direction between two (2) rows of the first contact structures **140**; and one (1) row of the second contact structures **141** is also interposed in the Y-direction between the two (2) rows of the first contact structures **140**. The one (1) row of the third contact structures **165** may be substantially aligned in the Y-direction with the one (1) row of the second contact structures **141**. In addition, in some embodiments, within horizontal boundaries of one of the staircase structures **116**, individual third contact structures **165** are interposed in the X-direction between two (2) columns each individually including two (2) first contact structures **140** and one (1) second contact structure **141** between the two (2) first contact structures **140**. In additional embodiments, quantities and/or an arrangement the first contact structures **140**, the second contact structures **141** (if any), and the third contact structures **165** are different than the quantities and/or the arrangement thereof depicted in FIG. 2C as described in further detail below with reference to FIGS. 3A-3C and FIGS. 4A-4C.

[0038] Thus, in accordance with embodiments of the disclosure, a microelectronic device comprises a stack structure, contact structures, and additional contact structures. The stack structure comprises a vertically alternating sequence of conductive material and insulative material arranged in tiers. The stack structure is divided into blocks each comprising a stadium structure including steps comprising horizontal ends of the tiers. The contact structures are within a horizontal area of the stadium structure and vertically extend through the stack structure. The additional contact structures are on at least some of the steps of the stadium structure and are coupled to the contact structures.

[0039] FIG. 3A shows a simplified, partial plan view depicting a different contact configuration within the portion A of the microelectronic device structure **100** shown in FIG. 1, in accordance with additional embodiments of the disclosure. As shown in FIG. 3A the horizontal positions of the first contact structures **140** may be shifted relative to the horizontal positions shown in FIG. 3C. For example, each of the first contact structures **140** may individually be substantially horizontally confined within a horizontal area of one of the steps **118**. Each of first contact structures **140** may be horizontally offset in the X-direction from each of the second contact structures **141**. Columns individually including some (e.g., two) of the first contact structures **140** may not be collinear with any of the second contact structures **141**. The arrangement shown in FIG. 3A may provide for greater separation between the first contact structures **140** and the second contact structures **141**. The arrangement may also provide for reduced separation between the first contact structures **140** and the third contact structures **165** relative to the arrangement shown in FIG. 2C. Such reduced separation may reduce a length of the associated routing structures **146** between individual first

contact structures **140** and individual third contact structures **165**.

[0040] FIG. 3B shows a simplified, partial plan view depicting a different contact configuration within the portion A of the microelectronic device structure **100** shown in FIG. 1, in accordance with additional embodiments of the disclosure. As shown in FIG. 3B, the staircase structure **116** may include first contact structures **140** but may be free of second contact structures **141** (FIG. 2C). Columns of the first contact structures **140** horizontal extending in parallel in the Y-direction may horizontally alternate in the X-direction with the third contact structures **165**, and each column may individually include four (4) of the first contact structures **140**. In addition, rows of the first contact structures **140** horizontally extending in parallel the X-direction may each be horizontally offset in the Y-direction from a row of the third contact structures **165** horizontally extending in the X-direction. In additional embodiments, each column of the first contact structures **140** includes a different quantity (e.g., less than four, greater than four) of the first contact structures **140** and/or one or more of the first contact structures **140** is replaced with one or more second contact structures **141** (FIG. 2C).

[0041] FIG. 3C shows a simplified, partial plan view depicting a different contact configuration within the portion A of the microelectronic device structure **100** shown in FIG. 1, in accordance with additional embodiments of the disclosure. As shown in FIG. 3C, there may be a relatively reduced number of the first contact structures **140** and a relatively increased number of the second contact structures **141** as compared to the configuration previously described with reference to FIG. 2C. In some embodiments, positions of at least some of the first contact structures **140** and second contact structures **141** within horizontally neighboring (e.g., in the X-direct) columns of the first contact structures **140** and the second contact structures **141** are different than one another. For example, horizontally neighboring columns of the first contact structures **140** and the second contact structures **141** may have first contact structures **140** positioned proximate different slot structures **112** (and, hence, different sides of an associated block **110** (FIG. 1)) than one another. At least some rows (e.g., rows proximate the slot structures **112**) of the first contact structures **140** and the second contact structures **141** extending in the X-direction may include an alternating sequence of the first contact structures **140** and the second contact structures **141** thereof.

[0042] FIGS. 4A, 4B, and 4C are simplified, partial plan views depicting different contact and routing configurations within the portion A of the microelectronic device structure shown in FIG. 1.

[0043] Referring to FIG. 4A, in some embodiments, the routing structures **146** are formed to horizontally extend between the third contact structures **165** (and the second additional contact structures **125** associated therewith) and the first contact structures **140** (and the first additional contact structures **124** associated therewith) within a single (e.g., only one) row of the first contact structures **140** positioned proximate one of the slot structures **112** (and, hence, one side of the block **110** (FIG. 1)). Each third contact structure **165** may individually be coupled to one (1) first contact structure **140** by way of an individual routing structure **146** (and an individual first additional contact structure **124** and an individual second additional contact structure **125**). The routing structures **146** may each exhibit substantially the same horizontal orientation. For example, each routing structure **146** may exhibit substantially the same diagonal orientation within the XY-plane. In additional embodiments, one or more of the routing structures **146** may be coupled to one or more of the first contact structures **140** within a different row of the first contact structures **140**, and/or one or more of the routing structures **146** may exhibit a different horizontal orientation than one or more other of the routing structures **146**.

[0044] Referring to FIG. 4B, in additional embodiments, the routing structures **146** are formed to horizontally extend between the third contact structures **165** (and the second additional contact structures **125** associated therewith) and the first contact structures **140** (and the first additional contact structures **124** associated therewith) within a single (e.g., only one) row of the first contact structures **140** relatively centrally positioned in the Y-direction within staircase structure **116** (and, hence, within the block **110** (FIG. 1) including the staircase structure **116**). Such a configuration

may minimize pathing distances of the routing structures **146** to couple the third contact structures **165** to the first contact structures **140**. The routing structures **146** may each exhibit substantially the same horizontal orientation. For example, each routing structure **146** may extend in the X-direction between the third contact structures **165** and the first contact structures **140** associated therewith. [0045] Referring to FIG. 4C, in further embodiments, the routing structures **146** are formed to horizontally extend between the third contact structures **165** (and the second additional contact structures **125** associated therewith) and the first contact structures **140** (and the first additional contact structures **124** associated therewith) within multiple (e.g., two) rows of the first contact structures **140** extending in the X-direction. For example, for each third contact structures **165**, at least two (2) routing structures **146** may be coupled thereto, the routing structures **146** horizontally extending from the third contact structure **165** to first contact structures **140** within different rows of the first contact structures **140** than one another. In some embodiments, the different rows of the first contact structures **140** are positioned proximate different slot structures **112** (and, hence, proximate different sides of the block **110** (FIG. 1)) than one another. In additional embodiments, one or more of the routing structures **146** may be coupled to one or more of the first contact structures **140** within a different row of the first contact structures **140**, such as a relatively centrally positioned row (e.g., in the Y-direction) of the first contact structures **140**. Coupling multiple routing structures **146** within an individual third contact structure **165** facilitates redundancy in design. Other variations are envisioned. An individual third contact structure **165** may be electronically connected to a desired quantity of first contact structures **140** through the routing structures **146** (and the first additional contacts structures **124** and the second additional contact structures).

[0046] Microelectronic device structures in accordance with embodiments of the disclosure may be included in microelectronic devices of the disclosure. For example, in accordance with embodiments of the disclosure, FIG. 5 illustrates a partial cutaway perspective view of a portion of a microelectronic device **501** (e.g., a memory device, such as a 3D NAND Flash memory device) including a microelectronic device structure **500**. The microelectronic device structure **500** may be substantially similar to one of the microelectronic device structures **100** previously described with reference to FIGS. 1, 2A-2C, 3A-3C, and 4A-4C. For clarity and ease of understanding the drawings and associated description, some features (e.g., structures, materials) of the microelectronic device structures **100** previously described herein are not shown in FIG. 5. However, it will be understood that any features of the microelectronic device structures **100** previously described with reference to one or more of FIGS. 1, 2A-2C, 3A-3C, and 4A-4C may be included in the microelectronic device structure **500** of the microelectronic device **501** described herein with reference to FIG. 5.

[0047] As shown in FIG. 5, the microelectronic device structure **500** of the microelectronic device **501** may include first contact structures **540** (corresponding to the first contact structures **140** previously described herein) coupled to third contact structures **565** (corresponding to the third contact structures **165** previously described herein) by way of routing structures **564** (corresponding to the routing structures **146** previously described herein). The first contact structures **540** may be positioned within a horizontal area of stadium structures **514** (corresponding to the stadium structures **114** previously described herein) within blocks (corresponding to the blocks **110** previously described herein) of a stack structure **502** (corresponding to the stack structure **102** previously described herein) separated from one another by slot structures **512** (corresponding to the slot structures **112** previously described herein). The first contact structures **540** may vertically extend through the stack structure **502** to conductive structures **560** (e.g., conductive island structures) of a source tier **559** underlying the stack structure **502**. The conductive structures **560** may be coupled to a base structure **568** including at least one control logic region having control logic devices configured to control various operations of other features (e.g., strings of memory cells **554**) of the microelectronic device **501**. The inclusion of first contact structures **540** within the

horizontal area of the stadium structures **514** may reduce the number of first contact structures **540** within crest regions (e.g., corresponding to the crest regions **122** previously described herein) between the horizontally neighboring (e.g., in the X-direction) stadium structures **514**.

[0048] As shown in FIG. 5, in addition to the features of the microelectronic device structure **500** previously described herein in relation to one or more of the microelectronic device structures **100** (FIGS. 1, 2A-2C, 3A-3C, and 4A-4C), the microelectronic device **501** may further include cell pillar structures **552** vertically extending through individual blocks of the stack structure **502**. The cell pillar structures **552** may be positioned within regions (e.g., memory array regions) of the blocks horizontally offset (e.g., in the X direction) from the stadium structures **514** within the blocks. Intersections of the cell pillar structures **552** and the conductive material **534** of the tiers **508** of the stack structure **502** form strings of memory cells **554** vertically extending through individual blocks of the stack structure **502**. For each string of memory cells **554**, the memory cells **554** thereof may be coupled in series with one another. Within an individual block, the conductive material **534** of some of the tiers **508** of the stack structure **502** may serve as access line structures (e.g., word line structures) for the strings of memory cells **554** within the block. In some embodiments, within each block, the memory cells **554** formed at the intersections of the conductive material **534** of some of the tiers **508** and the cell pillar structures **552** comprise so-called “MONOS” (metal-oxide-nitride-oxide-semiconductor) memory cells. In additional embodiments, the memory cells **554** comprise so-called “TANOS” (tantalum nitride-aluminum oxide-nitride-oxide-semiconductor) memory cells, or so-called “BETANOS” (band/barrier engineered TANOS) memory cells, each of which are subsets of MONOS memory cells. In further embodiments, the memory cells **554** comprise so-called “floating gate” memory cells including floating gates (e.g., metallic floating gates) as charge storage structures. The floating gates may horizontally intervene between central structures of the cell pillar structures **552** and the conductive material **534** of the different tiers **508** of the stack structure **502**.

[0049] The microelectronic device **501** may further include first select gates **556** (e.g., upper select gates, drain select gates (SGDs)), select line routing structures **566**, one or more second select gates **558** (e.g., lower select gates, source select gate (SGSs)), and digit line structures **562**. The digit line structures **562** may vertically overlie and be coupled to the cell pillar structures **552** (and, hence, the strings of memory cells **554**). At least one source structure within source tier **559** may vertically underlie and be coupled to the cell pillar structures **552** (and, hence, the strings of memory cells **554**). In addition, different third contact structures **565** (e.g., some third contact structures **565A** coupled to the conductive material **534** of the tiers **508** and the some of the first contact structures **540**; some other third contact structures **565B** coupled to the first select gates **556** and the some other of the first contact structures **540**) may couple various features of the microelectronic device **501** to one another as shown.

[0050] The control logic region of the base structure **568** may include one or more (e.g., each) of charge pumps (e.g., VCCP charge pumps, VNEGWL charge pumps, DVC2 charge pumps), delay-locked loop (DLL) circuitry (e.g., ring oscillators), Vdd regulators, drivers (e.g., string drivers), page buffers, decoders (e.g., local deck decoders, column decoders, row decoders), sense amplifiers (e.g., equalization (EQ) amplifiers, isolation (ISO) amplifiers, NMOS sense amplifiers (NSAs), PMOS sense amplifiers (PSAs)), repair circuitry (e.g., column repair circuitry, row repair circuitry), I/O devices (e.g., local I/O devices), memory test devices, MUX, error checking and correction (ECC) devices, self-refresh/wear leveling devices, and other chip/deck control circuitry. The control logic region of the base structure **568** may be coupled to various conductive structures (e.g., at least some of the conductive structures **560**, at least one source structure) of the source tier **559**, the routing structures **564**, the select line routing structures **566**, and the digit line structures **562**. In some embodiments, the control logic region of the base structure **568** includes CMOS (complementary metal-oxide-semiconductor) circuitry. In such embodiments, the control logic region of the base structure **568** may be characterized as having a “CMOS under Array” (“CuA”)

configuration.

[0051] Thus, in accordance with embodiments of the disclosure, a memory device comprises a stack structure, first contact structures, second contact structures, routing structures, and strings of memory cells. The stack structure comprises a vertically alternating sequence of conductive material and insulative material arranged in tiers. The stack structure has blocks separated from one another by dielectric slot structures. Each of the blocks comprises stadium structures individually comprising opposing staircase structures each having steps comprising edges of the tiers of the stack structure. The first contact structures are within horizontal boundaries of the opposing staircase structures of at least some of the stadium structures. The first contact structures vertically extend through the stack structure to conductive structures below the stack structure. The second contact structures are on the steps of at least one of the opposing staircase structures of each of the at least some of the stadium structures. The routing structures horizontally extend between and couple the first contact structures to the second contact structures. The strings of memory cells vertically extend through the stack structure and are coupled to the second contact structures.

[0052] Microelectronic devices structures (e.g., the microelectronic device structures **100** previously described with reference to FIGS. **1**, **2A-2C**, **3A-3C**, and **4A-4C**) and microelectronic devices (e.g., the microelectronic device **501** previously described with reference to FIG. **5**) in accordance with embodiments of the disclosure may be used in embodiments of electronic systems of the disclosure. For example, FIG. **6** is a block diagram of an illustrative electronic system **600** according to embodiments of disclosure. The electronic system **600** may comprise, for example, a computer or computer hardware component, a server or other networking hardware component, a cellular telephone, a digital camera, a personal digital assistant (PDA), portable media (e.g., music) player, a Wi-Fi or cellular-enabled tablet such as, for example, an iPad® or SURFACE® tablet, an electronic book, a navigation device, etc. The electronic system **600** includes at least one memory device **602**. The memory device **602** may comprise, for example, one or more of a microelectronic device structure (e.g., one of the microelectronic device structures **100** previously described with reference to FIGS. **1**, **2A-2C**, **3A-3C**, and **4A-4C**) and a microelectronic device (e.g., the microelectronic device **501**) previously described herein. The electronic system **600** may further include at least one electronic signal processor device **604** (often referred to as a “microprocessor”). The electronic signal processor device **604** may, optionally, include one or more of a microelectronic device structure (e.g., one of the microelectronic device structures **100** previously described with reference to FIGS. **1**, **2A-2C**, **3A-3C**, and **4A-4C**) and a microelectronic device (e.g., the microelectronic device **501**) previously described herein. While the memory device **602** and the electronic signal processor device **604** are depicted as two (2) separate devices in FIG. **6**, in additional embodiments, a single (e.g., only one) memory/processor device having the functionalities of the memory device **602** and the electronic signal processor device **604** is included in the electronic system **600**. In such embodiments, the memory/processor device may include one or more of a microelectronic device structure (e.g., one of the microelectronic device structures **100** previously described with reference to FIGS. **1**, **2A-2C**, **3A-3C**, and **4A-4C**) and a microelectronic device (e.g., the microelectronic device **501**) previously described herein. The electronic system **600** may further include one or more input devices **606** for inputting information into the electronic system **600** by a user, such as, for example, a mouse or other pointing device, a keyboard, a touchpad, a button, or a control panel. The electronic system **600** may further include one or more output devices **608** for outputting information (e.g., visual or audio output) to a user such as, for example, a monitor, a display, a printer, an audio output jack, a speaker, etc. In some embodiments, the input device **606** and the output device **608** comprise a single touchscreen device that can be used both to input information to the electronic system **600** and to output visual information to a user. The input device **606** and the output device **608** may communicate electrically with one or more of the memory device **602** and the electronic signal processor device **604**.

[0053] Thus, in accordance with embodiments of the disclosure, an electronic system comprises an

input device, an output device, a processor device operably connected to the input device and the output device, and a memory device operably connected to the processor device. The memory device comprises a stack structure, first contact structures, second contact structures, third contact structures, and control logic circuitry. The stack structure comprises a vertically alternating sequence of conductive material and insulative material arranged in tiers. The stack structure comprises stadium structures having steps comprising portions of the conductive material of the tiers. The first contact structures vertically extend through the stack structure and are within horizontal areas of the stadium structures. The second contact structures vertically extend through the stack structure and are within the horizontal areas of the stadium structures. The third contact structures are on the steps of the stadium structures. The third contact structures are in electrical communication with the first contact structures and are electrically isolated from the second contact structures. The control logic circuitry underlies the stack structure and is in electrical communication with the first contact structures.

[0054] The structures, devices, and systems of the disclosure advantageously facilitate one or more of improved performance, reduced costs (e.g., manufacturing costs, material costs), increased miniaturization of components, and greater packaging density as compared to conventional structures, conventional devices, and conventional systems. For example, the inclusion of first contact structures (e.g., the first contact structures **140, 540**) of the disclosure within horizontal boundaries of the stadium structures (e.g., the stadium structures **114, 514**) of the disclosure may reduce number of first contact structures needed within the crest region (e.g., the crest regions **122**) interposed between the stadium structures as compared to conventional designs. As a result, the crest regions of the disclosure may have reduced horizontal dimensions relative to conventional designs. The relatively reduced sizes of the crest regions of the disclosure may facilitate a relatively greater density of memory cells within a microelectronic device (e.g., the microelectronic device **501**) of the disclosure as compared to conventional microelectronic devices, and/or may facilitate relatively reduced dimensions overall for the microelectronic device of the disclosure relative to conventional microelectronic devices. The structures, devices, and systems of the disclosure may also improve scalability, efficiency, and simplicity as compared to conventional structures, conventional devices, and conventional systems.

[0055] While the disclosure is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and have been described in detail herein. However, the disclosure is not limited to the particular forms disclosed. Rather, the disclosure is to cover all modifications, equivalents, and alternatives falling within the scope of the following appended claims and their legal equivalents. For example, elements and features disclosed in relation to one embodiment may be combined with elements and features disclosed in relation to other embodiments of the disclosure.

Claims

1. A microelectronic device, comprising: a stack structure comprising a vertically alternating sequence of conductive material and insulative material arranged in tiers, the stack structure divided into blocks individually comprising a stadium structure including steps comprising horizontal ends of the tiers; and contact structures individually within a horizontal area of the stadium structure of a respective one of the blocks, at least three of the contact structures within horizontally overlapping one of the steps of the stadium structure and comprising: a first contact vertically extending completely through the stack structure; a second contact coupled to the first contact and vertically extending partially through the stack structure and terminating at the step; and a third contact electrically isolated from the first contact and the second contact and vertically extending completely through the stack structure.
2. The microelectronic device of claim 1, wherein the second contact is relatively more proximate

to the third contact and relatively more distal from the first contact.

3. The microelectronic device of claim 1, wherein the second contact is offset from the third contact in a first horizontal direction and is substantially aligned with the third contact in a second horizontal direction orthogonal to the first horizontal direction.

4. The microelectronic device of claim 3, wherein the second contact is offset from the first contact in the first horizontal direction and in a second horizontal direction orthogonal to the first horizontal direction.

5. The microelectronic device of claim 3, wherein the first contact is substantially aligned with the third contact in the first horizontal direction and is offset from the third contact in a second horizontal direction orthogonal to the first horizontal direction.

6. The microelectronic device of claim 3, wherein the first contact is offset from the third contact in the first horizontal direction and in a second horizontal direction orthogonal to the first horizontal direction.

7. A memory device, comprising: a stack structure comprising a vertically alternating sequence of conductive material and insulative material arranged in tiers; a memory array region comprising strings of memory cells vertically extending through the stack structure; and a staircase region horizontally offset from the memory array region and comprising: stadium structures individually including steps defined by horizontal ends of the tiers of the stack structure; a first contact positioned at least partially within a horizontal area of one of the steps of one of the stadium structures; a second contact horizontally positioned entirely within the horizontal area of the one of the steps of the one of the stadium structures, the second contact coupled to the conductive material of one of the tiers of the stack structure; and a third contact horizontally positioned at least partially within the horizontal area of the one of the steps of the one of the stadium structures, the third contact electrically isolated from the first contact and the second contact.

8. The memory device of claim 7, wherein the second contact is coupled to the first contact.

9. The memory device of claim 7, wherein the second contact substantially horizontally centered within the horizontal area of the one of the steps of the one of the stadium structures.

10. The memory device of claim 7, wherein a horizontal center of the second contact is horizontally offset from horizontal centers of the first contact and the third contact in each of a first direction and a second direction orthogonal to the first direction.

11. The memory device of claim 7, wherein: the first contact is horizontally positioned entirely within the horizontal area of the one of the steps of the one of the stadium structures; and the third contact is horizontally positioned only partially within the horizontal area of the one of the steps of the one of the stadium structures.

12. The memory device of claim 7, wherein the first contact and the third contact are individually horizontally positioned only partially within the horizontal area of the one of the steps of the one of the stadium structures.

13. The memory device of claim 7, wherein the first contact is coupled to a control logic circuitry underlying the stack structure and a routing structure overlying the stack structure, the routing structure coupling the first contact to the second contact.

14. A microelectronic device, comprising: a stack structure comprising a vertically alternating sequence of conductive material and insulative material arranged in tiers, the stack structure partitioned into blocks respectively comprising: a staircase structure including steps defined by horizontal ends of the tiers; a first contact structure horizontally positioned at least partially within a horizontal area of one of the steps of the staircase structure and extending vertically through all of the tiers of the stack structure; a second contact structure horizontally positioned within the horizontal area of and vertically terminating at the one of the steps of the staircase, the second contact structure coupled to the first contact structure; and a third contact structure horizontally positioned at least partially within the horizontal area of the one of the steps of the staircase structure and extending vertically through all of the tiers of the stack structure, the third contact

structure electrically isolated from the second contact structure, and the second contact structure horizontally positioned relatively more proximate to the third contact structure than the first contact structure.

15. The microelectronic device of claim 14, wherein the third contact structure is horizontally positioned relatively more distal from a slot structure vertically extending completely through the stack structure than is the first contact structure.

16. The microelectronic device of claim 14, wherein the third contact structure is electrically isolated from the first contact structure.

17. The microelectronic device of claim 14, wherein the second contact structure vertically extends partially through the stack structure and terminates at the conductive material of one of the tiers of the stack structure.

18. The microelectronic device of claim 14, wherein the first contact structure and the third contact structure individually extend vertically through an entire vertical height of the stack structure.

19. The microelectronic device of claim 14, further comprising routing structures extending between and coupling the first contact structure to the second contact structure.

20. The microelectronic device of claim 14, further comprising control logic circuitry in electrical communication with the first contact structures.
